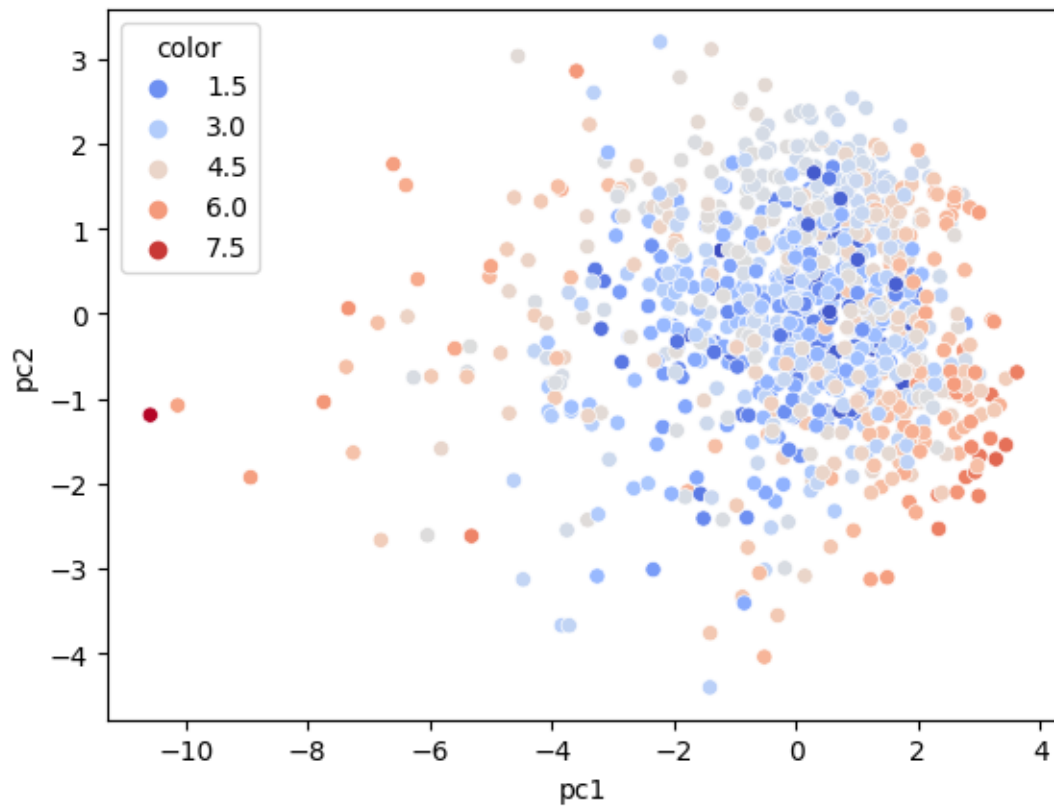


0.1 Question 2d

Create a 2D scatterplot of the first two principal components of `mid1_grades_centered_scaled`. Use `colorize_midterm_data` to add a color column to `mid1_1st_2_pcs`. Your code will be very similar to the code from problems 2a and 2b.

```
In [38]: u_2d, s_2d, vt_2d = np.linalg.svd(mid1_grades_centered_scaled.to_numpy(), full_matrices=False)
sns.scatterplot(data=colorize_midterm_data(pd.DataFrame(mid1_grades_centered_scaled.to_numpy()
```



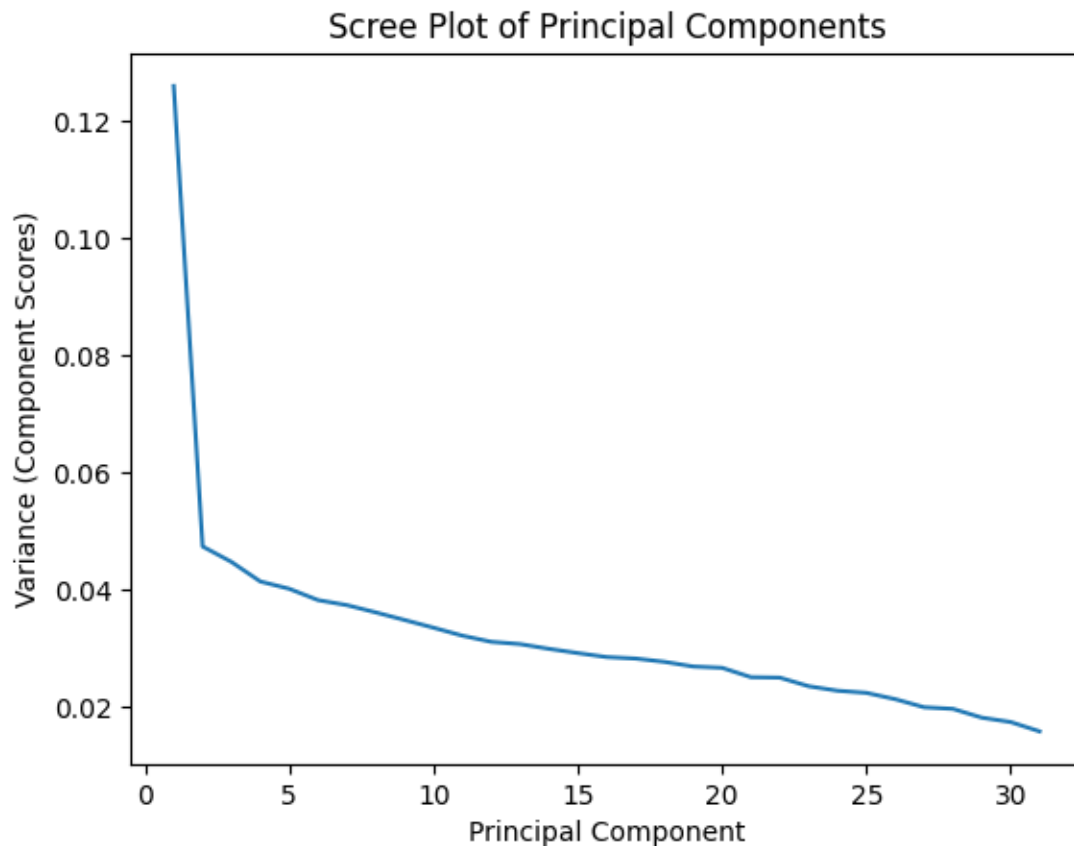
0.2 Question 2e

If you compute the fraction of the variance captured by this 2D scatter plot, you'll see it's only 17%, roughly 12% by the 1st PC, and roughly 5% by the 2nd PC. **In the cell below, create a scree plot showing the fraction of the variance explained by each principle component using the data from 2d.**

Informally, we can say that our midterm scores matrix has a high rank. More formally, we can say that a rank 2 approximation only captures a small fraction of the variance, and thus the data are not particularly amenable to 2D PCA scatterplotting.

```
In [39]: plt.xlabel("Principal Component")
         plt.ylabel("Variance (Component Scores)")
         plt.title("Scree Plot of Principal Components")
         plt.plot(range(1, len(s_2d) + 1), (s_2d ** 2) / sum(s_2d ** 2))
```

```
Out[39]: [<matplotlib.lines.Line2D at 0x2a0014910>]
```



Unfortunately, we have two problems:

1. There is a lot of overplotting, with only 27 distinct dots. This means that at least some states voted exactly alike in these elections.
2. We don't know which state is which because the points are unlabeled.

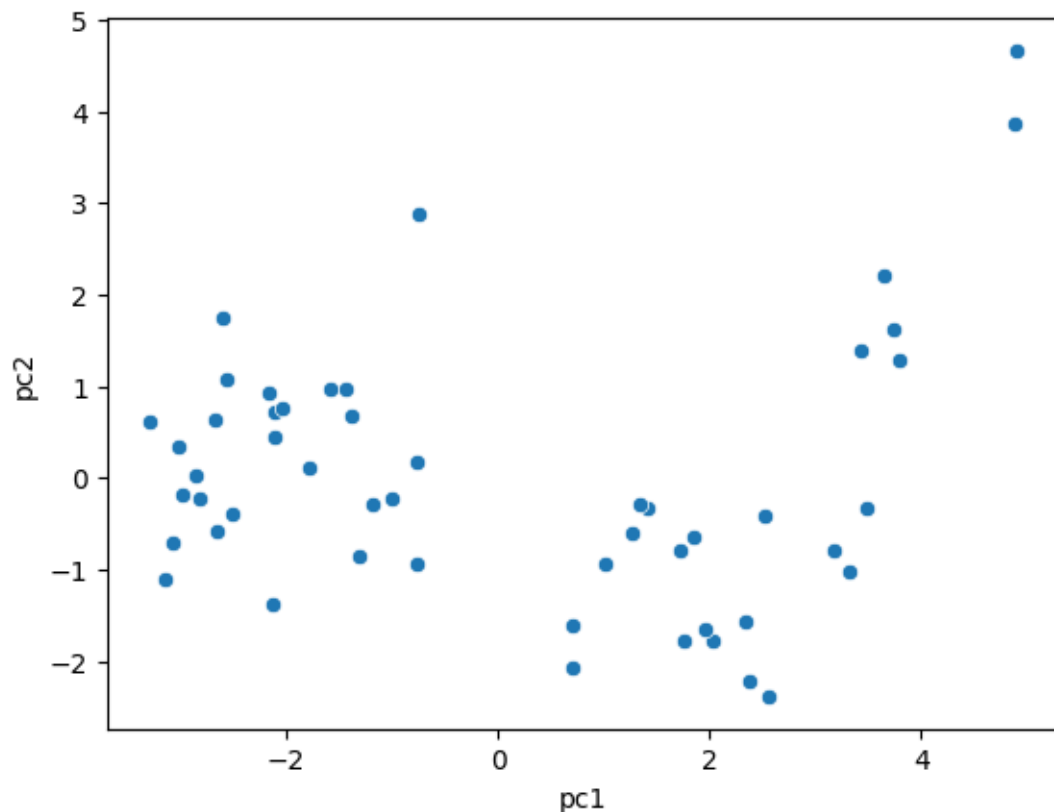
Let's start by addressing problem 1.

In the cell below, create a new dataframe `first_2_pcs_jittered` with a small amount of random noise added to each principal component. In this same cell, create a scatterplot.

The amount of noise you add should not significantly affect the appearance of the plot, it should simply serve to separate overlapping observations. Don't get caught up on the exact details of your noise generation, it's fine as long as your plot looks roughly the same as the original scatterplot.

Hint: See the pairplot from the intro to question 2 for an example of how to introduce noise.

```
In [51]: first_2_pcs_jittered = first_2_pcs + np.random.normal(0, 0.5, size=(len(first_2_pcs), 2))
sns.scatterplot(data = first_2_pcs_jittered, x = "pc1", y = "pc2");
```



Give an example of a cluster of states that vote a similar way. Does the composition of this cluster surprise you? If you're not familiar with U.S. politics, it's fine to just say 'No, I'm not surprised because I don't know anything about U.S. politics.'

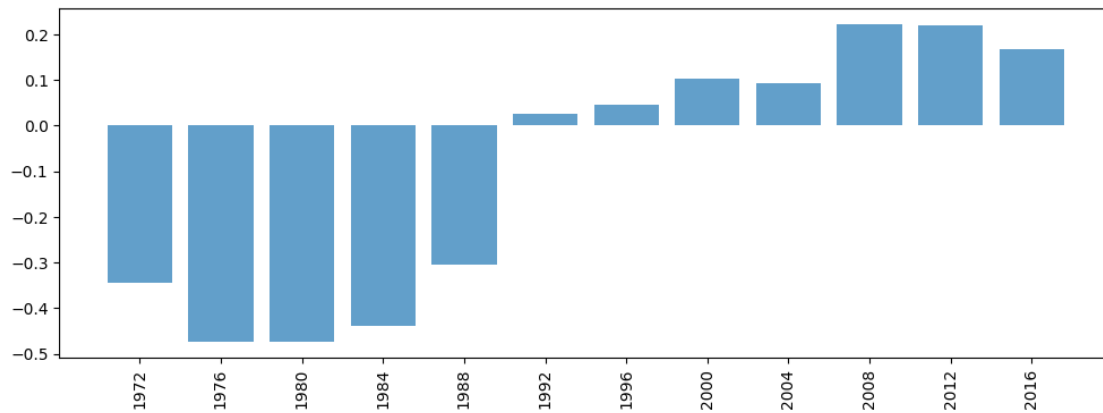
There is a cluster on the left hand side that votes in a similar way. Specifically there's an area of South Carolina, Alabama, Texas, and Kansas. Particularly interesting as they are geographically close to each other. Otherwise, in terms of politics the cluster doesn't surprise me as I don't know anything specific about how US politics works.

In the cell below, write down anything interesting that you observe by looking at this plot. You will get credit for this as long as you write something reasonable that you can take away from the plot.

It's interesting that if you were to do a split between the middle, you get two clusters. Could you possible classify them as the two different policial parties? The Democrat and Republicans. There also happens to be a third tiny cluster of only DC and Minnesota. Do they represent the third uncommon party of libertarians? Or are they simply outliers due to population density?

In the cell below, plot the the 2nd row of V^T .

```
In [54]: plt.figure(figsize=(12, 4))  
         plot_pc(list(df_1972_to_2016.columns), vt_q3, 1);
```



0.3 Question 3i

Using your plots from question 3h as well as the original table, give a description of what it means to have a relatively large positive value for **pc1** (right side of the 2D scatter plot), and what it means to have a relatively large positive value for **pc2** (top side of the 2D scatter plot).

In other words, what is generally true about a state with relatively large positive value for **pc1**? For a large positive value for **pc2**?

Note: **pc2** is pretty hard to interpret, and the staff doesn't really have a consensus on what it means either. We'll be nice when grading.

Note: Principal components beyond the first are often hard to interpret (but not always; see question 1 earlier in this homework).

Higher principal component values could correlate to a higher probability of a state voting for a certain party. If the pc value is small, might mean that there is less correlation. Low pc values also might mean that the state is less confident or has less probability in being a certain group.

0.4 Question 3j

To get a better sense of whether our 2D scatterplot captures the whole story, create a scree plot for this data. On the y-axis plot the fraction of the total variance captured by the i th principal component. You should see that the first two principal components capture much more of the variance than we were able to capture when using the Data 100 Midterm 1 data. It is partially for this reason that the 2D scatter plot was so much more useful for this dataset.

Hint: Your code will be very similar to the scree plot from problem 1d. Be sure to label your axes appropriately!

```
In [56]: plt.xlabel("Principal Component")
         plt.ylabel("Variance (Component Scores)")
         plt.title("Scree Plot of Principal Components")
         plt.plot(s_q3 ** 2 / u.shape[0])
```

```
Out[56]: [<matplotlib.lines.Line2D at 0x2a02da040>]
```

